Literature Review for Multimodal Student Wellbeing Classification

SWEM: Student Wellbeing Evaluation and Monitoring

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Abstract

This literature review contextualizes the machine-learning work implemented in the project's machine-learning folder, which is located in `Scripts/ML/` rather than a top-level `ML/` directory. The project uses window-level posture, blink, and heart-rate features to classify student wellbeing states under supervised and unsupervised learning settings. The reviewed literature covers affective and physiological computing, computer-vision-based posture and blink measurement, heart-rate variability, supervised tabular classification, clustering, anomaly detection, and evaluation methodology. The review also compares these methods with the implemented experiments, including direct and hierarchical three-class classification, Leave-One-Group-Out validation, sensor ablation, data augmentation, and participant-normalized unsupervised clustering.

1 Introduction

Student wellbeing monitoring sits at the intersection of affective computing, physiological computing, human-computer interaction, and applied machine learning. Affective computing frames emotional and cognitive-state inference as a computational problem using observable signals such as behavior, physiology, and interaction traces [16]. Physiological computing similarly studies how bodily signals can be used to infer user state, workload, fatigue, or stress [8, 35]. In the present project, the target application is not clinical diagnosis, but a proof-of-concept system for identifying study-session states from sensor-derived features [26].

The project, named SWEM (Student Wellbeing Evaluation and Monitoring), classifies a user's state as focused/normal, sleepy, or stressed using three biometrics: posture from webcam video, blinking from webcam facial landmarks, and heart-rate parameters from a photoplethysmography sensor or CSV file [26]. The machine-learning component implements both supervised classification and unsupervised analysis. The supervised pipeline evaluates binary normal-versus-abnormal classification, sleepy-versus-stress subtype classification, hierarchical three-class classification, and direct three-class classification [28]. The unsupervised pipeline evaluates binary and three-class clustering, anomaly detection, sensor ablation, augmentation, and participant-normalized clustering [23, 32, 34].

This review explains the academic background needed to understand the implemented work. It focuses on methods that are actually present in the repository: Logistic Regression, Support Vector Machines, Random Forests, Gradient Boosting, K-Means, Gaussian Mixture Models, Agglomerative Clustering, Isolation Forest, One-Class SVM, Local Outlier Factor, group-aware cross-validation, and common classification and clustering metrics. It also discusses the feature families used in the project: posture geometry, eye-blink dynamics, PERCLOS, heart rate, and RMSSD-like heart-rate variability.

2 Background

2.1 Problem Definition

The implemented machine-learning task uses 60-second windows as samples. Each row is a summary of posture, blinking, and heart-rate behavior over one window [29, 33]. The supervised data contain four CSV files: `normal.csv`, `anomalous.csv`, `sleepy.csv`, and `stress.csv`. The main binary task is normal versus anomalous. The main three-class task is normal, sleepy, and stress. The result summary verifies that `anomalous.csv` is the union of `sleepy.csv` and `stress.csv` after relabeling, so the direct three-class experiment correctly excludes `anomalous.csv` to avoid duplicate samples with incompatible labels [29].

The final supervised dataset has 649 window-level samples for the binary task and 649 samples for the direct three-class task. The three-class dataset contains 218 normal samples, 213 sleepy samples, and 218 stress samples. The four participant/source groups are Alessia, Alvaro, Beatrice, and Mo [29]. The group column used for evaluation is `video_source`. This small number of participants is an important limitation because the effective population size is much smaller than the row count, especially because overlapping windows are correlated.

2.2 Feature Representation

The project uses 18 numeric features. Nine posture features summarize shoulder asymmetry, head lateral tilt, and a forward-head-posture proxy using mean, standard deviation, and bad-percentage statistics. Seven blink features summarize eye aspect ratio, PERCLOS, blink count, blink frequency, and inter-blink-interval statistics. Two heart features, `HR` and `RMMSD`, represent heart rate and an RMSSD-like heart-rate-variability measurement [21, 24, 28].

Computer-vision feature extraction is based on landmark estimation. MediaPipe provides a general perception-pipeline framework [13]; the project documentation states that posture extraction uses MediaPipe Pose landmarks, consistent with the BlazePose family of real-time pose-estimation methods [1, 24]. Eye-blink features are based on the Eye Aspect Ratio (EAR), a facial-landmark measure that has been used for real-time blink detection [19]; the broader MediaPipe Face Mesh literature is also relevant to facial-landmark-based eye analysis [10]. PERCLOS, the percentage of time that the eyes are closed, is commonly used as an alertness or fatigue-related indicator [6]. The project documentation gives threshold bands for EAR and PERCLOS, but the exact bibliographic source for those threshold bands is not fully specified in the repository.

The posture documentation reports a 7-degree shoulder-asymmetry threshold, a 10-degree head-lateral-tilt threshold, and a forward-head-proxy threshold of 0.7 [24, 25]. The posture review file lists links to scoliosis-screening, CUELA ergonomic-norm, and webcam-posture resources, but several entries are not complete bibliographic records.

For heart-rate variability, the repository notes that the CSV column is named `RMMSD`, while the scientifically common term is RMSSD [29]. RMSSD is a standard time-domain heart-rate-variability index based on successive differences between beat intervals [35].

2.3 Learning Settings

The supervised component trains labeled classifiers. It compares Logistic Regression, Support Vector Machines, Random Forests, and Gradient Boosting [28]. Logistic Regression provides a linear probabilistic baseline; SVMs provide margin-based nonlinear classification through kernels [4]; Random Forests combine many randomized decision trees and are strong baselines for nonlinear tabular problems [2]; Gradient Boosting builds additive tree ensembles by sequentially improving a differentiable loss approximation [7].

The unsupervised component fits clustering and novelty/anomaly models without labels during fitting. The clustering methods include K-Means [14], Gaussian Mixture Models estimated

with expectation-maximization [5], and Agglomerative Clustering, historically associated with hierarchical clustering procedures such as Ward's method [37]. The anomaly-detection methods include Isolation Forest [12], One-Class SVM [18], and Local Outlier Factor [3]. Labels are used afterward to evaluate cluster alignment and map arbitrary cluster IDs to semantic classes [30, 33].

2.4 Evaluation

The supervised implementation uses Leave-One-Group-Out cross-validation, holding out one participant/source at a time [28, 29]. This is appropriate because overlapping windows and participant-specific baselines can make random row-level splitting too optimistic. General cross-validation principles are well established in model evaluation [11]; the project adapts this idea by using participant/source groups as folds.

The supervised scripts report accuracy, balanced accuracy, macro F1, weighted F1, precision, recall, classification reports, and confusion matrices [28]. The unsupervised scripts report clustering metrics such as adjusted Rand index, normalized mutual information, V-measure, silhouette score, Davies-Bouldin score, and Calinski-Harabasz score, plus majority-vote mapped classification metrics [32, 34]. Silhouette analysis is a classic internal cluster-validity method [17]; adjusted Rand index and information-theoretic clustering measures are widely used for external cluster validation [9, 36].

3 Related Work

3.1 Affective and Physiological Computing

Affective computing established the broader idea that computational systems can sense, model, and respond to human affective states [16]. Later physiological-computing work showed that sensor signals can support stress-state inference in practical settings, including real-world driving tasks [8]. The SWEM project follows this general pattern: it does not infer wellbeing from a single signal, but combines posture, eye behavior, and heart-rate features [26]. Compared with much affective-computing literature, the present implementation is narrower and more application-specific: it targets student study sessions and uses engineered window-level features rather than end-to-end raw-signal modeling.

3.2 Vision-Based Posture Monitoring

Pose-estimation frameworks make it possible to convert ordinary RGB video into body-landmark trajectories. MediaPipe and BlazePose are relevant because the posture module extracts landmark-based quantities such as shoulder asymmetry, head tilt, and a forward-head proxy [1, 13, 24]. The project's posture review also motivates threshold-based posture features using scoliosis-screening and ergonomic-norm references [25]. However, because those notes do not provide complete BibTeX-quality records for every cited threshold, the present review treats those thresholds as project-documented design choices rather than fully validated universal thresholds.

This feature-engineering approach is appropriate for a small dataset because it reduces high-dimensional video data to interpretable low-dimensional descriptors. Its limitation is that the thresholds may depend on camera placement, user anatomy, sitting position, and landmark-detection quality. A lateral or multi-camera setup would measure forward-head posture more directly than a frontal-only proxy.

3.3 Blink and Fatigue Indicators

Eye-blink monitoring is commonly based on facial landmarks or image-based eye-state classification. The Eye Aspect Ratio is a compact geometric feature derived from eye landmarks and has been used for real-time blink detection [19]. PERCLOS is an established alertness-related measure that summarizes eye closure over time [6]. The SWEM blinking module follows this general literature by computing window-level EAR statistics, blink counts, blink frequency, inter-blink intervals, and PERCLOS [21]. This makes the blink representation interpretable and compatible with tabular classifiers.

The limitation is that blink behavior is sensitive to lighting, camera angle, glasses, occlusion, and individual differences. The project partially addresses this through validity checks and window aggregation, but the available machine-learning files do not report a separate landmark-quality ablation.

3.4 Heart Rate and Heart-Rate Variability

Heart-rate variability is a standard family of physiological measures. RMSSD, the root mean square of successive differences between beat intervals, is one recognized time-domain measure [35]. In the SWEM dataset, the feature is stored as `RMMSD`, and the result summaries explicitly note that this corresponds scientifically to RMSSD [29, 33]. Physiological stress-detection work has shown that physiological sensors can help distinguish stress-related states [8]. In the present project, sensor ablation indicates that heart features are especially important for separating sleepy from stress, while posture is more important for normal-versus-abnormal classification [22, 27].

3.5 Classical Machine Learning for Small Tabular Sensor Datasets

The implemented supervised models are standard choices for small to medium tabular datasets. Logistic Regression provides a simple linear baseline. Support Vector Machines can model nonlinear boundaries with kernels [4]. Random Forests are robust nonlinear ensembles and often perform well on engineered tabular features [2]. Gradient Boosting is another tree-ensemble family that builds additive models in a sequential fashion [7]. The repository uses scikit-learn pipelines for imputation, scaling where needed, and model fitting [15, 28].

For unsupervised learning, the project uses both clustering and anomaly detection. K-Means is a centroid-based baseline [14]; Gaussian Mixture Models provide probabilistic soft clustering with covariance modeling [5]; Agglomerative Clustering provides a hierarchical clustering baseline [37]. Isolation Forest, One-Class SVM, and Local Outlier Factor represent different anomaly-detection assumptions: isolation by random partitioning, boundary learning around normal data, and local-density deviation respectively [3, 12, 18].

4 Methodological Context

4.1 Supervised Direct and Hierarchical Classification

The supervised implementation has two methods for three-class classification. The direct method trains one classifier to predict normal, sleepy, or stress. The hierarchical method trains a binary normal-versus-abnormal model and a second sleepy-versus-stress subtype model; the subtype model is used only when the binary model predicts abnormal [28, 29]. This hierarchical design matches the label structure because `anomalous` is the union of sleepy and stress.

The result summary reports that Random Forest is the selected model for binary classification, subtype classification, direct three-class classification, and both stages of the best hierarchical classifier [29]. The best hierarchical model reaches macro F1 of approximately 0.957, while the best direct three-class Random Forest reaches macro F1 of approximately 0.951 [29].

This difference is small but suggests that the two-stage decomposition is useful on the available data.

The methodological advantage of the hierarchical approach is interpretability: it separates abnormality detection from subtype identification. Its limitation is error propagation. If the binary stage predicts a stress or sleepy sample as normal, the subtype model is never applied. The direct model avoids this gate but has to solve the full three-class problem at once.

4.2 Preprocessing and Pipelines

The supervised code uses `SimpleImputer(strategy="median")` for missing values. Logistic Regression and SVM also use `StandardScaler`; Random Forest and Gradient Boosting use imputation without scaling [28]. The unsupervised code uses median imputation and standard scaling because distance-based and covariance-based models are sensitive to feature scale [32, 34].

Using scikit-learn pipelines is important because imputation and scaling are learned inside each training fold rather than on the full dataset [15, 28]. This reduces leakage from the held-out participant into training.

4.3 Evaluation and Model Selection

The supervised implementation ranks models by macro F1, then balanced accuracy, weighted F1, and accuracy [29]. Macro F1 is appropriate because the project cares about each wellbeing class rather than only the majority class. The binary dataset has 431 anomalous samples and 218 normal samples, so accuracy alone could hide class-specific errors [29].

The unsupervised scripts distinguish fitting from evaluation. Clustering is performed without labels, but labels are used afterward for majority-vote cluster mapping and external metrics [30, 33]. This is standard practice when cluster IDs are arbitrary, but it is not the same as having a deployable unsupervised classifier with semantic labels.

4.4 Sensor Ablation, Augmentation, and Participant Normalization

Sensor ablation tests the contribution of posture, blink, and heart features. The supervised ablation report shows that removing heart features strongly hurts subtype, hierarchical, and direct three-class classification, while binary normal-versus-abnormal performance changes only slightly [22, 27]. This indicates that heart-rate and HRV features are central to sleepy-versus-stress separation in the current data.

The supervised augmentation experiment applies Gaussian jitter only inside each Leave-One-Group-Out training fold and never augments the held-out participant [20]. It slightly improves binary, hierarchical, and direct three-class macro F1, but the summary recommends reporting augmentation carefully because the dataset has only four participants [20]. The unsupervised augmentation results improve clustering but worsen LOSO anomaly detection [31]. This difference suggests that augmentation can stabilize cluster structure while still harming novelty-detection boundaries.

Participant-normalized unsupervised clustering transforms each feature relative to a participant's normal baseline [23]. This strongly improves unsupervised results, especially z-score normalization, but it assumes that a normal calibration segment is available for each participant. This is a methodological strength for personalized systems and a limitation for immediate deployment.

5 Comparison with Existing Literature

The project is similar to affective and physiological computing literature in that it uses multimodal signals to infer user state [8, 16]. It is similar to landmark-based computer-vision work

because it converts video into interpretable body and eye features using pose and face landmarks [1, 13, 19]. It is similar to HRV-based physiological analysis because it includes HR and an RMSSD-like feature [35].

The project differs from many broader affective-computing systems in its deliberately compact engineered-feature design. It does not train a deep model on raw video or raw physiological signals. Instead, it uses 18 window-level features and classical machine-learning models [28, 32]. This is a reasonable choice for the small dataset because classical models generally require fewer examples than end-to-end deep architectures.

The project's main implementation contribution is not a new algorithm, but a complete applied comparison across learning settings and sensor conditions. It compares direct and hierarchical three-class classification, supervised and unsupervised models, clean and masked missing-sensor scenarios, augmentation, and participant-normalized clustering [22, 23, 27, 29, 31]. These comparisons make the work useful as an implementation study of multimodal student-state classification.

6 Gaps and Limitations

6.1 Limitations in the Literature Context

The broader literature supports the use of posture, eye behavior, and heart-rate variability as user-state signals [6, 8, 16, 19, 35]. However, several gaps remain relevant to this project. First, physiological and behavioral indicators are context-dependent; models trained in controlled settings may not transfer to different users, devices, lighting conditions, or study environments.

Second, webcam-based frontal posture measures are indirect for some anatomical dimensions. The project's forward-head proxy is scale-normalized, but frontal video cannot directly measure anterior-posterior head displacement as reliably as a side view or multi-sensor setup.

Third, unsupervised clusters can align with labels but do not automatically provide semantic class names. Majority-vote mapping is valid for evaluation, but deployment would require calibration labels, expert rules, or another interpretation mechanism [9, 33, 36].

6.2 Limitations of the Implemented Work

The main limitation of the implemented work is data scale. The result summaries repeatedly state that the dataset contains four participants and that results should be treated as proof of concept rather than strong evidence of generalization [29, 30, 33]. Leave-One-Group-Out validation is the correct internal evaluation strategy, but four folds cannot cover the diversity of real students.

Another limitation is sensor synchronization. The machine-learning files consume already merged CSVs, while the dataset-building code outside `Scripts/ML/` aligns some heart-rate data by row position rather than precise timestamps. The literature review therefore treats HR/RMSSD findings as project-specific rather than a general physiological claim.

The perfect sleepy-versus-stress Random Forest result also needs cautious interpretation [29]. Sensor ablation shows that heart features largely drive this separation [22, 27]. This may represent meaningful physiological differences in the collected data, but it may also reflect recording conditions or participant-specific patterns. External validation is needed.

Finally, the participant-normalized unsupervised results are strong, but they assume a normal calibration segment for each participant [23]. That assumption is realistic for personalized systems but not for cold-start deployment.

7 Conclusion

The literature supports the general direction of the SWEM project: multimodal behavioral and physiological features can be used for user-state inference [8, 16]; pose and face landmarks can provide interpretable computer-vision features [1, 13, 19]; HRV is a recognized physiological signal family [35]; and classical supervised and unsupervised models are appropriate baselines for small engineered-feature datasets [2–5, 12, 14].

Within this context, the implemented work is best understood as a proof-of-concept multimodal student-wellbeing classification study. Its strongest supervised result is the hierarchical Random Forest three-class model, and its unsupervised results show meaningful but weaker alignment between feature structure and wellbeing labels [29, 30, 33]. The review also identifies the main limitations that should be addressed before stronger claims are made: more participants, more precise sensor synchronization, verified literature citations for project-specific thresholds, external validation, and careful handling of personalized calibration assumptions.

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